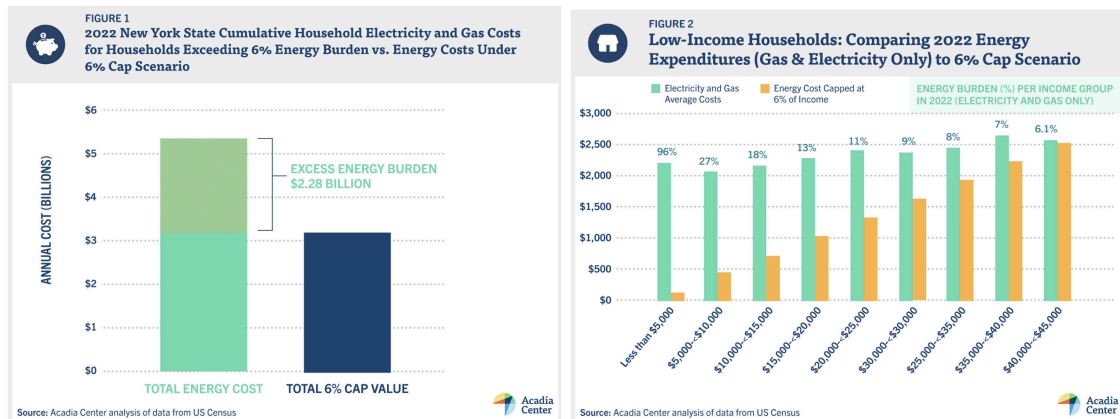


Introduction & background

Energy is foundational to modern urban life, powering systems ranging from public infrastructure and transportation to essential household functions such as heating, cooling, lighting, and appliances. As cities grow and energy demand increases, access to affordable and reliable energy has become a critical determinant of quality of life. This importance is highlighted by global policy initiatives like the United Nations' Sustainable Development Goals, which emphasize universal access to affordable and reliable energy services by 2030 (United Nations, 2023).

However, around the world's major cities a growing share of households are energy insecure or burdened. In 2020 New York City, **where the comptroller defined a high energy burden as spending more than 6% of income on utility costs and a severe burden above 10%**, reported that 25% of households (approx. 1,850,000) surpassed the 6% threshold (ACEEE, 2020). With the median energy burden for low-income households being 9.3%, three times higher than the citywide median of 2.9%; with some households reaching the 17% mark (ACEEE, 2020). Across demographic groups, 32% Black households and 46% Hispanic households were affected, reflecting a disparity rooted in historical patterns of disinvestment and housing inequality (Carley & Konisky, 2020).

More broadly, across New York State, over 2 million households exceed the 6% threshold, collectively incurring \$2.28 billion in excess energy costs in 2022 alone, with the burden falling disproportionately on low-income households earning less than \$45,000 per year (Acadia Center, 2025). This financial burden can be seen in the figures below:



Energy burden is not only an issue of grid efficiency but most importantly one of social and environmental justice. High energy burden has been linked to health outcomes, increased financial stress, and housing instability, primarily among low income households who often live in older less energy efficient units (Hernández and Bird., 2010). These households also have a limited capacity to adapt to price fluctuations, intensifying economic vulnerability and demonstrating why understanding the drivers of energy burden is crucial to develop public policies that address both sustainability and equity.

Prior research has identified two primary categories of predictors in residential energy use: urban form, including housing density, building age, compactness, land-use mix, and access to transit and greenspace; and socioeconomic/demographic characteristics, such as income, tenure, and household size (Güneralp et al., 2017; Ewing & Rong, 2008; Ratti et al., 2005). While socioeconomic variables are consistently reported as strong predictors, the role played by urban form is less clear, particularly within city scale. In places like NYC where built form has a significant variation across boroughs and neighborhoods, understanding the contributions of physical characteristics can be insightful for the development of both policies and infrastructure interventions.

With this study we aim to address this gap by applying machine learning methods to predict household energy burden at the census tract level across the city. Trying to answer the question: ***To what extent do urban form characteristics improve the prediction of household energy burden beyond socioeconomic factors alone?***

Literature Review

Prior research on residential energy consumption and burden highlights the importance of both **urban form** and **socioeconomic characteristics** in shaping energy outcomes. The Intergovernmental Panel on Climate Change (IPCC) has found that urban form contributes significantly to energy consumption, influencing both building performance and transportation demand. Gunerlap et al. (2017) found that higher urban density can reduce per-household energy consumption at levels comparable to energy efficiency improvements, as denser environments lower heating and cooling loads while reducing vehicle miles traveled.

Socioeconomic factors are equally significant, low-income households face disproportionately high energy burdens relative to income, while renters are further disadvantaged by the split-incentive problem, where landlords bear the costs of efficiency updates while tenants pay utility bills (Hernandez & Bird, 2010; Melvin, 2018).

Several machine learning models have been applied to model energy consumption and burden. Among those decision trees and random forest are particularly effective in capturing nonlinear relationships between the variables. Chun et al. (2025) used decision trees to predict energy burden across the United States, finding no single dominant predictor; instead, burden is shaped by localized combinations of environmental, social, and economic conditions.

Unsupervised learning methods have been widely used to analyze energy patterns. K-means clustering has been used to segment residential energy consumption patterns and support policy targeting (Kim et al., 2026). Building on this, Zhang et al. (2015) showed that clustering-based methods can outperform traditional formulaic approaches in predicting baseline energy demand when demographic and

usage data are combined. However, while these studies apply clustering to analyze energy loads, most focus on large-scale datasets or direct consumption rather than affordability. As Chun et al. (2025) found, addressing energy burden requires a tailored approach, as regional environmental, social, and economic conditions shape burden in ways that aggregate models may miss. Our research builds on this literature by applying these methods in a granular, city-specific context to inform energy policy in New York City.

However, while previous studies apply decision tree models and clustering to analyze energy loads, most of this work focuses on large scale data sets or direct energy consumption. As Chun et al. found, addressing energy burden requires a tailored approach as regional factors including environmental, social, and economic conditions influence energy burden. Our research seeks to build upon the literature by utilizing these machine learning methods in a granular city and burden specific context, in order to improve energy policy in New York City.

Methodology & Modeling Results

Data framework and preprocessing

For this study urban form and socioeconomic factors are used to model energy burden. We integrate multiple datasets that capture both variables at the census tract level in New York City. The target variable, energy burden (defined as the percentage of household income spent on utilities), is obtained from the Department of Energy's Low-Income Energy Affordability Data (LEAD) Tool.

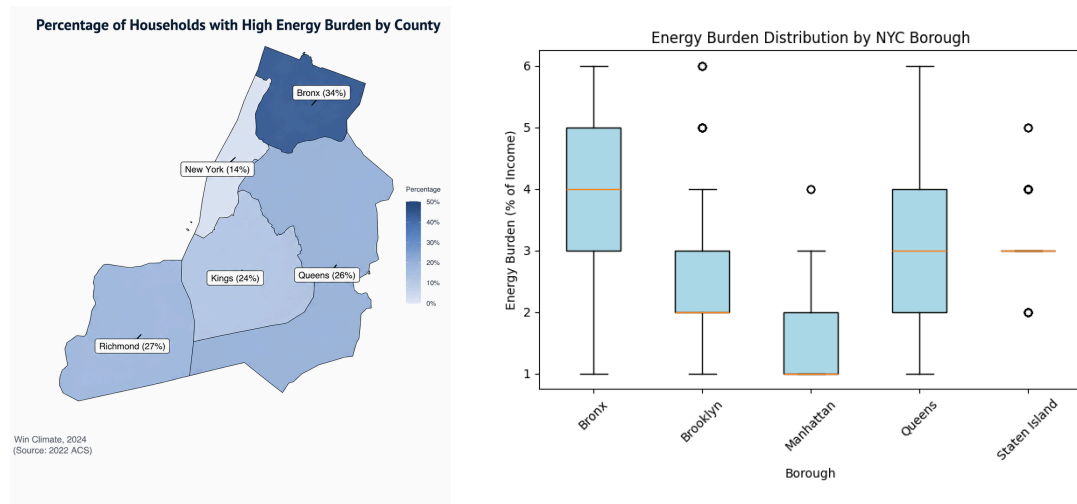
Urban form variables were derived from the NYC PLUTO dataset and include building age, floor area ratio (FAR), building size, and number of floors. The NYC Street Tree Census was used to approximate tree canopy coverage and associated cooling effects, and MTA subway and bus stop data to compute transit accessibility metrics. Socioeconomic variables are from the American Community Survey (ACS) 5-Year Estimates 2020-2024 and include median household income, tenure (percent renters), racial composition, household size, and housing burden indicators. The Climate and Economic Justice Screening Tool (CEJST) dataset was used for validation and contextual comparison. Appendix 1 contains further details on the datasets and extracted features from each.

All datasets were merged using the census tract level, which required normalization across different spatial units. Prior to modeling the data underwent preprocessing to ensure consistency and quality. Missing values were handled using median imputation, and invalid placeholder values were identified and removed. Non-residential tracts with zero energy burden are excluded from the analysis. The final data frame contains 37 features and more than 2,200 census tracts.

Features were organized into two groups: urban form variables (building age, effective year, floor area ratio, building and residential area per unit, lot area, commercial share, residential share, historic and landmark designations, tree canopy coverage, temperature differential, and transit stop counts) and

socioeconomic variables (median income, percent non-white, percent renters, average household size, percent rent-burdened, and total households). The full variables and datasets lists are provided in the Appendix.

Prior to analysis, the spatial distribution of energy burden across New York City boroughs was analyzed. This step was necessary to validate that the data was reasonably representative, evenly distributed, and ensure that the dataset aligns with patterns documented in the existing literature. The findings show that the Bronx has the highest percentage of households experiencing high energy burden, followed by Staten Island and Queens, confirming that our data sets follow reported and documented burden trends.



The figures above highlight that energy burden patterns are not uniformly distributed across boroughs. Instead, they reveal clear spatial disparities and significant variability within boroughs. These findings support our hypothesis that energy burden is shaped by a combination of spatial factors, built environment characteristics, and socioeconomic conditions. To further investigate these relationships, we develop an analytical approach that integrates both supervised and unsupervised machine learning models.

Modeling & Results: Random Forest Ablation

To better understand the drivers of household energy burden across New York City, a Random Forest regression model was implemented. This approach was selected due to its ability to capture nonlinear relationships and complex interactions between variables that traditional linear models may overlook. It is robust to outliers and multicollinearity, and produces feature importance scores that directly address the research question about which characteristics most influence energy burden.

The dataset was split into 80% training and 20% test sets. To quantify the independent contribution of urban form, we conduct an ablation analysis comparing three models:

1. A Linear Regression model using only socioeconomic variables
2. A Random Forest model using only socioeconomic variables
3. A full Random Forest model including both socioeconomic and urban form variables

Initially establishing a baseline model using linear regression, including only socioeconomic variables, allowed us to quantify how much of the variation in energy burden could be explained by these alone, providing a simple benchmark for comparison. We then trained a simple Random forest model using the same variables to test whether a nonlinear model could better capture variation in energy burden; this model will also be used as a baseline for the full random forest. For the two baseline models the results were: **Linear Regression (socioeconomic only):** RMSE = 0.92451, $R^2 = 0.4653$. **Random Forest (socioeconomic only):** RMSE = 0.8723, $R^2 = 0.5386$

These results show that the Random Forest model outperforms linear regression even when using the same set of socioeconomic variables. This suggests that the relationship between socioeconomic characteristics and energy burden is not purely linear, and that nonlinear interactions between variables play a meaningful role in explaining variation in energy burden across census tracts. Reinforcing the need for a more flexible modeling approach.

For the full random forest, hyperparameters were tuned for using RandomizedSearchCV with 3-fold cross-validation, exploring variations in the number of estimators, maximum tree depth, minimum samples per leaf, and maximum features considered at each split. Model performance is evaluated using 5-fold cross-validation on the training set and validated on the held-out test set using metrics including R^2 and RMSE.

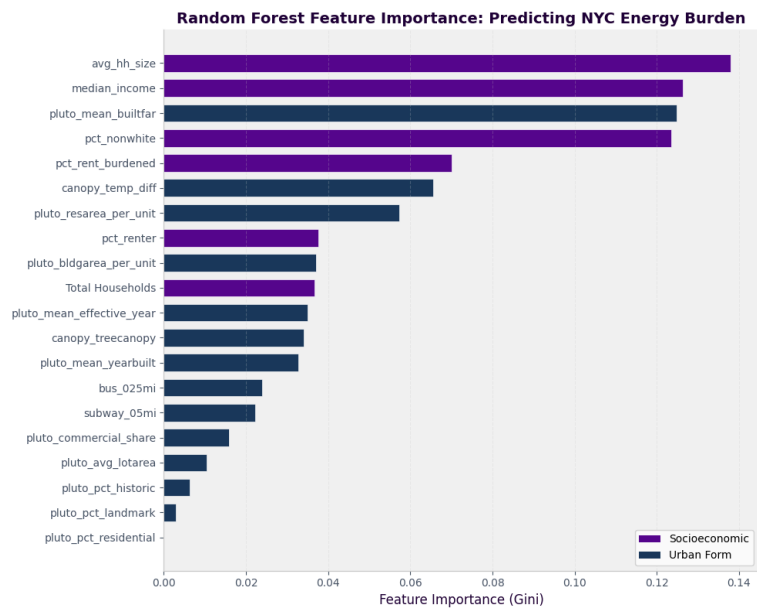
Model	RMSE	R^2
Linear Regression (socioeconomic only)	0.9251	0.4653
Random forest (socioeconomic only)	0.8723	0.5244
Random forest Tuned (socioeconomic only)	0.8677	0.5296
Random forest (Full)	0.8229	0.5760
Random forest Tuned (Full)	0.8183	0.5817

Prior to the tuning, the baseline Random Forest, which used 200 trees, achieved an **RMSE: 0.8229** and **R^2 : 0.5760** when incorporating the full set of features. In comparison, the socioeconomic only model achieved a lower R^2 , resulting in a difference of **ΔR^2 of 0.0515** between models. This difference represents the independent contribution of urban form variables in explaining variation in

energy burden. After tuning, the optimized Random Forest model had 300 estimators, a maximum depth of 15 and minimum of 2 samples per leaf, achieving an improved performance of **RMSE = 0.8183** and **$R^2 = 0.5817$** , making it the best performing model overall.

Hyperparameter tuning produced only marginal gains over the baseline Random Forest ($\Delta R^2 = 0.0057$), suggesting the added computational cost may not be warranted. The consistent performance gap between full and socioeconomic-only models ($\Delta R^2 = 0.05$) confirms that urban form contributes meaningfully to explaining energy burden.

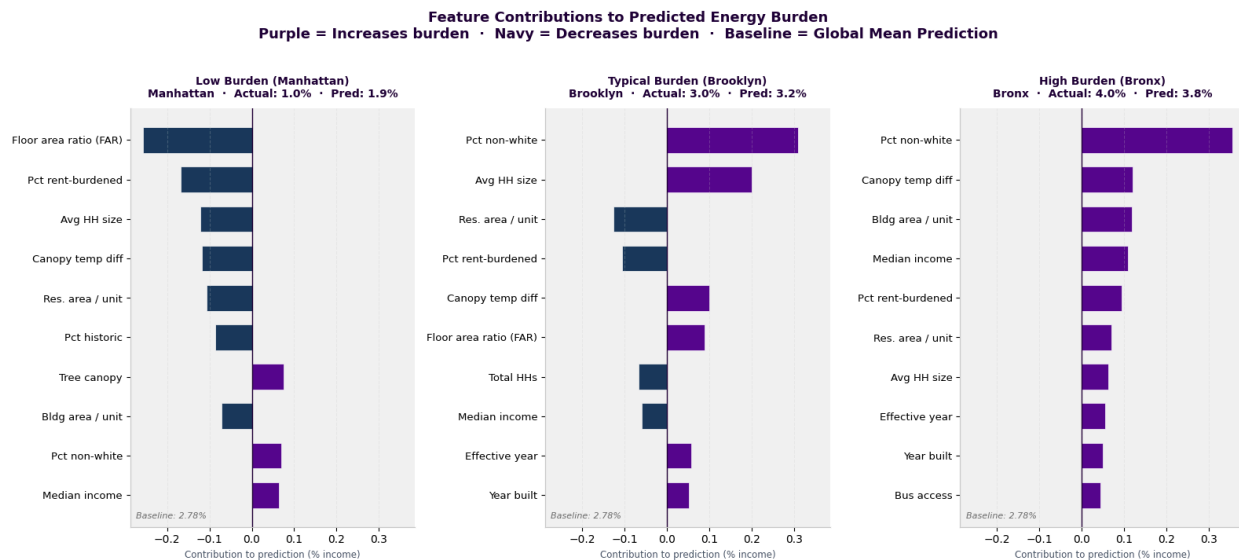
Furthermore we conducted a feature analysis on the whole set, shown below, which provided an insight into which variables drive burden rates. Socioeconomic variables, particularly average household size, median income, and percent non-white population, emerged as the most influential predictors, aligning with findings by literature that place economic capacity as a primary determinant of energy affordability. However, these results must be interpreted in context: according to the 2020 U.S.



Census and subsequent estimates, nearly 70% of New York City's population identifies as non-white, with especially high shares in boroughs such as Brooklyn (65%) and the Bronx (91%). This raises an important consideration, as the strong predictive power of the percent non-white variable should not be understood as causal, but rather as a proxy for broader structural inequalities, including historical disinvestment, disparities in housing quality, and unequal access to resources.

In the comparative analysis below, we examine how features contribute to predicted energy burden in three illustrative census tracts: a low-burden tract in Manhattan, a typical-burden tract in Brooklyn, and a high-burden tract in the Bronx. In the Manhattan tract, urban form variables such as floor area ratio pull the prediction well below the citywide baseline, suggesting that density and building efficiency are playing a protective role independent of income. The Brooklyn tract shows a more mixed profile, with socioeconomic and urban form features partially offsetting each other. In the Bronx tract, nearly all contributing features push the burden upward, including both socioeconomic variables and

urban form characteristics such as heat exposure and building area per unit, with no meaningful downward offset from the built environment. This tract-level decomposition illustrates how urban form can either mitigate or compound socioeconomic stress depending on local conditions.



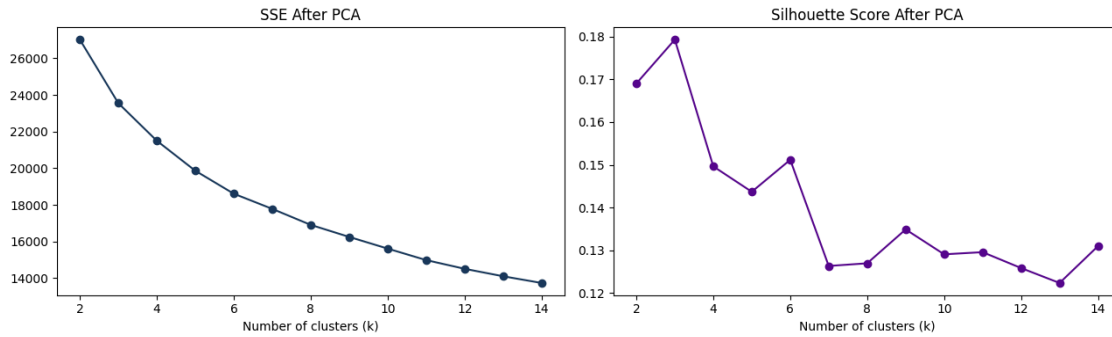
This reinforces the idea that race-related variables capture broader patterns of systemic inequity, while the built environment acts as a mechanism through which these inequalities are materially expressed. In high-burden areas like the Bronx, the combination of socioeconomic disadvantage and less efficient or more exposed urban form amplifies energy costs, highlighting that addressing energy burden requires both social and physical interventions within the urban system.

Modeling & Results: Clustering

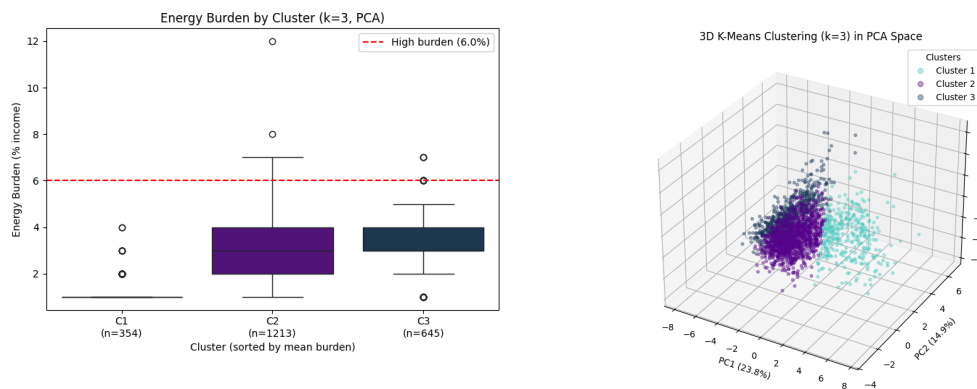
The random forest analysis was followed by four clustering models. Both K-means and gaussian mixture clustering produced reliable clusters. However, hierarchical clustering did not produce reliable results, as most data points fell into one cluster. In single and complete link clustering, cluster 1 had 2,211 values, and cluster 2 had 1 value. Given these results, K-means was selected for the analysis, as interpretability should be prioritized since the results will be used to shape policymaking. While gaussian mixture provides probabilistic cluster assignments, making the results more uncertain therefore less suitable. Additionally, K-means better captures the separation between energy burdened and non-burdened households.

To handle potential multicollinearity and reduce the dimensionality, principal component analysis (PCA) was performed. Which determined that 10 components were required to capture 90% of the total variance, effectively condensing our features without losing significant information.

The clustering results were validated by analyzing the changes in SSE (Sum of Squared Errors) and Silhouette scores. Model's robustness was assessed using the Calinski-Harabasz Score, which measures the ratio of between-cluster to within-cluster dispersion (where a higher score is preferred), and the Davies-Bouldin Index which assesses cluster compactness (where a lower score is preferred). Based on the data, $k=3$ yields the highest Silhouette Score, while $k=2$ provides the highest Calinski-Harabasz Index. The Davies-Bouldin Index continues to decrease as k increases, which is expected as clusters become more specific. This comparison can be seen in the table below:



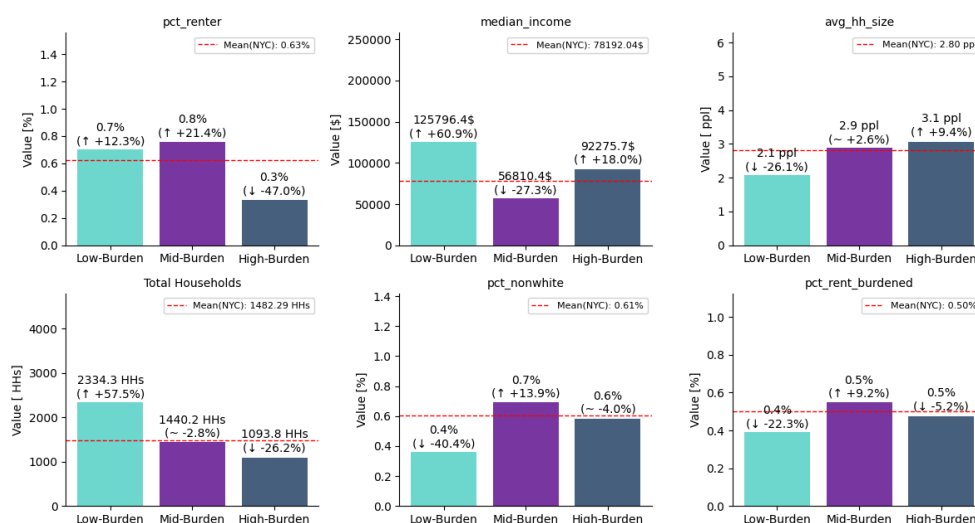
The K-Means clustering ($k=3$) reveals distinct energy burden profiles across NYC boroughs. Cluster 1 ($n=355$) represents the lowest burden, with 1.17% of energy making up income, and is dominated by Manhattan residents (63.4%). Cluster 2 ($n=1,200$) is the largest size, with a 2.97% of energy making up income, and a high proportion of Brooklyn residents (43.8%). Cluster 3 ($n=657$) denotes the highest average burden at 3.37%, with over half the group residing in Queens (51.3%).



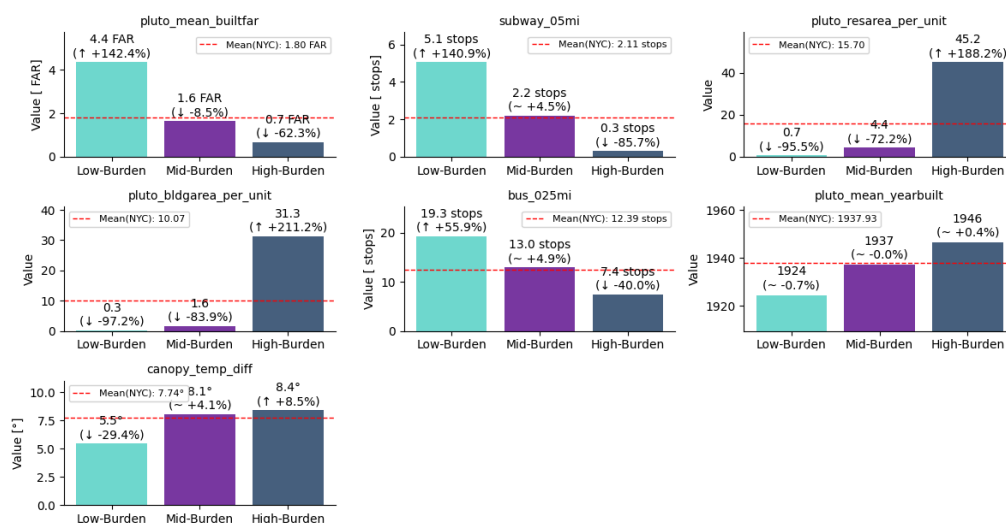
To better understand how socioeconomic and urban-form impacts each cluster, we compared the mean values of the three clusters against the NYC average. The low-burden cluster highlights higher median incomes and a greater number of total households than the other clusters. Conversely, the high-burden cluster shows lower percentages of renters and smaller total household counts.

In the analysis of urban form features, the low-burden cluster indicates higher floor-area ratio (FAR) values, more nearby transportation stops, older buildings, and smaller air temperature differences related to tree canopy. The high-burden cluster shows the opposite trend: lower FAR, fewer transportation stops, newer buildings, and higher temperature differences. Additionally, this cluster exhibits significantly higher scores for both residential area per unit and building area per unit.

Socio features analysis: Cluster vs. NYC Average



Non-Socio Features Analysis: Cluster vs. NYC Average



Building age was a top indicator across random forest and clustering. However, the top features for random forest were mostly dominated by the socioeconomic category, whereas the top features for K-means were dominated by the urban form category. Unlike random forest, we did not use top predictors for K-means as our form of analysis for understanding indicators for energy burden. Instead, we used the

average of variables and compared them across groups since listing the top variables would produce skewed results; variables that measure similar statistics would cluster together. We found that the variables with the largest differences between clusters were FAR values, proximity to transportation stops, building age, and air temperature differences related to tree canopy. The top predictors using random forest were average household size, median household income, FAR, percentage of non-white, and percentage of rent-burdened.

Discussion

Our findings across both models align closely with the existing literature, reinforcing the idea that energy burden is shaped by both socioeconomic conditions and characteristics of the built environment. Prior research demonstrates that electricity consumption can be predicted based solely on building characteristics alone (Papadopoulos et al., 2018), supporting our results that building characteristics help to explain energy burden. In particular, variables related to building density and structure, such as floor area ratio (FAR), are strongly linked to heat transfer and energy use, as higher surface area-to-volume ratios increase exposure to external temperatures and energy demand.

Although our dataset did not explicitly distinguish between single-family and multifamily housing, several variables serve as useful proxies. For instance, household size may indicate housing type, as larger households are more likely to occupy single-family homes, which typically consume more energy due to greater exposure to external conditions. Similarly, building age can reflect differences in construction typologies and energy efficiency. Our findings suggest that older buildings, particularly those in denser, multifamily configurations, may be associated with lower energy burden, consistent with research showing that older NYC housing stock can exhibit relatively efficient thermal performance (Furman center, 2022).

Socioeconomic variables remain the strongest predictors of energy burden. Median income, rent burden, and demographic composition consistently explain a large share of variation, reflecting broader structural inequalities in access to efficient housing and energy resources. This relates to findings done by Ko et al. who found that building age, HVAC systems, occupant income, and whether the occupant owns the unit also impact energy usage (Ko et al., 2013). In random forest, median income was the 2nd top predictor, and percentage of rent burden was the 5th top predictor of energy burden. Additionally, Ma and Cheng found that income is tied to housing quality, and those who are lower income tend to live in less energy efficient buildings (Ma and Cheng, 2016).

At the same time, urban form variables, such as proximity to transportation, tree canopy, and urban density, capture important environmental and spatial dynamics. For example, areas with greater

access to transit tend to be denser and more energy-efficient, while increased tree canopy and lower temperature differences help mitigate cooling demand, reducing overall energy costs.

In random forest, median income was the 2nd top predictor, and percentage of rent burden was the 5th top predictor of energy burden. Additionally, Ma and Cheng found that income is tied to housing quality, and those who are lower income tend to live in less energy efficient buildings (Ma and Cheng, 2016).

Ewig and Rong found that residential energy use is impacted by urban density since less dense counties are more likely to be single-family homes and therefore have higher energy use. This can be seen in our K-means results. Proximity to transportation is another way of measuring an area's density. Greater access to public transportation indicates greater urban density. The low-burden cluster had more nearby transportation stops compared to the mid-burden and high-burden clusters.

Our results for air temperature differences related to tree canopy are backed by Ko et al's findings that greenspace and city compactness reduce energy usage (Ko et al., 2013). Compactness, how tightly infrastructure is placed, can enhance natural ventilation and cool a city's microclimate. A study by McPherson et al. found that a 10% increase in tree cover in Chicago reduced heating and cooling costs by \$50-\$90 per unit.

Top predictors were not consistent across random forest and K-means modeling since each method handles data differently. Random forest is a form of supervised learning that finds top predictors using gini importance, and therefore provides feature weights. In contrast, K-means is a form of unsupervised learning and weighs all features equally, instead, evaluating on Euclidean distance from cluster centers. K-means is used to understand underlying data patterns, whereas random forest is used to predict a target variable. Since our data was labeled, using both methods helped to increase robustness checks on our analysis to understand NYC patterns in energy burden (K-means), and what variables are most influential in determining energy burden (random forest). Urban form factors grouped in K-means explain differences in clustering since urban form is relatively static and structural. This is also a reflection of a city's geography; dense areas with greater access to transportation typically have lower energy burden. In comparison, the results from random forest explain which features predict energy burden rather than structural factors. In our random forest results, socioeconomic conditions explained the greatest variance in energy burden, likely due to the fact that income, race, and whether someone is a renter fluctuates greatly across census tracts compared to transportation access. For instance, several residential buildings may have the same value for proximity to transportation, but within those buildings, the socioeconomic features vary widely.

Recommendations

As said, our findings reinforce existing evidence that socially vulnerable populations are disproportionately affected by energy burden. Current NYC policies, such as Local Laws 84 and 97, aim to improve energy efficiency and reduce emissions, while programs like LIHEAP provide financial assistance to households experiencing energy poverty. However, our results suggest that additional interventions are needed.

Short-term strategies should focus on direct financial relief, including expanded energy vouchers and increased enrollment in demand response programs, which can help households manage peak energy costs. These approaches provide immediate support for vulnerable populations without requiring structural changes.

At the same time, long-term solutions must address the built environment. Investments in building retrofits, insulation improvements, and energy-efficient systems can reduce structural energy demand, particularly in older or poorly maintained housing. Additionally, expanding urban green infrastructure, such as tree canopy, can mitigate urban heat island effects and lower cooling costs. While these interventions require greater time and investment, they are essential for achieving sustained reductions in energy burden.

Limitations

While our research uses 10 features to analyze energy burden, other variables in the literature include building material, building orientation and configuration, and aspect ratio. Due to limitations in data availability and time constraints, we were unable to include these variables in our analysis. Further research should include these variables to get a more accurate prediction of energy burden. Additionally, the focus of our study was limited to predicting energy burden at the census tract level due to the Department of Energy lack of granularity. A more comprehensive understanding of energy burden at the household level would help target certain policies to address the multifaceted aspects of energy burden.

Our research is also limited to the biases in the LEAD energy burden statistics. Calculations of energy burden vary across studies. Replication of metered utility data would help strengthen the validity of burden estimates, which currently rely on modeled rather than observed utility expenditures

As a final note, results for Staten Island should be interpreted with caution, as the borough is significantly less populated than others in NYC. This lower population density may lead to sampling bias and reduced representativeness, potentially affecting the reliability of model outputs and clustering results for this area.

Conclusion

Approximately 25% of NYC households are energy burdened, spending more than 6% of their income goes on energy-related costs. This has implications for urban livability, especially in a city consistently ranked as having one of the highest costs of living. We sought to understand what factors contribute to energy burden using random forest to find features that predict energy burden, and K-means modeling to understand NYC's structural patterns in energy burden.

This analysis found that the top socioeconomic indicators of energy burden in NYC are median household income, percentage of non-white, and percentage of being rent burdened. The top urban form indicators are FAR, proximity to transportation, air temperature differences, and building age. Our research indicates that energy burden is a highly interconnected issue within urban systems, impacted by poverty, environmental justice, and accessibility.

Therefore, there are several approaches to addressing energy burden, which have the co-benefits of increasing living affordability, greenspaces, and access to transportation. Together, these findings suggest that while socioeconomic disadvantage is the primary driver of energy burden, urban form plays a meaningful and measurable secondary role, particularly in high-burden areas like the Bronx where poor building conditions and limited transit access compound economic vulnerability. Addressing the energy burden in NYC therefore requires a dual approach: targeted income and subsidy policies for immediate relief, and longer-term physical interventions in the built environment to reduce structural energy costs for the most vulnerable communities to promote more equitable urban living conditions.

Contribution of each team member

Zoe Tseng: Data cleaning, Clustering validation and analysis, report writing

Eunji Kim: Data cleaning, Clustering analysis, report draft, slides

Isabella Vera: Data cleaning, RF tuning and analysis, report writing

Kelsey Knight: Data cleaning, RF modeling and analysis, report writing, slides

Appendix 1: Dataset references

Dataset	Description	Reference (APA)
DOE LEAD Tool	Provides modeled estimates of household energy burden at the census tract level, including energy expenditures as a percentage of income. Used as the primary dependent variable in this study.	U.S. Department of Energy. (2026). <i>Low-Income Energy Affordability Data (LEAD) Tool</i>. https://energy.gov/scep/slsc/lead-tool
NYC PLUTO	Contains parcel-level land use and building characteristics, including building age, floor area ratio (FAR), and gross floor area. Used to derive urban form variables and proxies for housing structure and density.	New York City Department of City Planning. (2026). <i>Primary Land Use Tax Lot Output (PLUTO)</i>. https://data.cityofnewyork.us/City-Government/Primary-Land-Use-Tax-Lot-Output-PLUTO-/64uk-42ks
NYC Street Tree Census (Tree Equity / Canopy Proxy)	Includes detailed information on street trees such as location, species, and condition. Used to approximate tree canopy coverage and analyze its relationship to urban heat and energy burden.	New York City Department of Parks & Recreation. (2015). <i>2015 Street Tree Census</i>. https://data.cityofnewyork.us/Environment/2015-Street-Tree-Census-Tree-Data/uvpi-gqn timer
MTA Subway Stations & Bus Stops	Used to calculate proximity-to-transit variables as a proxy for urban density and accessibility.	Metropolitan Transportation Authority. (2026.). <i>MTA Subway Stations</i>. https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f Metropolitan Transportation Authority. (n.d.). <i>MTA Bus Stops</i>. https://data.ny.gov/Transportation/MTA-Bus-Stops/2ucp-7wg5
ACS 5-Year Estimates	Supplies tract-level socioeconomic and housing data, including median income, household size, tenure (owner vs. renter), and racial composition. Used to construct key demographic predictors.	U.S. Census Bureau. (n.d.). <i>American Community Survey 5-Year Estimates</i>. https://data.census.gov

Appendix 2: feature definitions

```
feature_cols = [
    'pluto_mean_yearbuilt', 'pluto_mean_effective_year', 'pluto_mean_builtfar',
    'pluto_bldgarea_per_unit', 'pluto_resarea_per_unit',
    'pluto_avg_lotarea', 'pluto_commercial_share', 'pluto_pct_residential',
    'pluto_pct_historic', 'pluto_pct_landmark',
    'canopy_treecanopy', 'canopy_temp_diff',
    'subway_05mi', 'bus_025mi',
    'median_income', 'pct_nonwhite', 'pct_renter', 'avg_hh_size',
```

'pct_rent_burdened', 'Total Households']

socio_cols = [

'median_income', 'pct_nonwhite', 'pct_renter',

'avg_hh_size', 'pct_rent_burdened', 'Total Households'

]

References

- Acadia Center. (2025). New York's household energy burden: Imperative challenges and solutions. <https://acadiacenter.org/resource/new-yorks-household-energy-burden-imperative-challenges-and-solutions/>
- American Council for an Energy-Efficient Economy (ACEEE). (2020). Energy burden. <https://www.aceee.org/energy-burden>
- Barr, J. M. (2016). Building the skyline: The birth and growth of Manhattan's skyscrapers. Oxford University Press. <https://buildingtheskyline.org/housing-twenties/>
- Carley, S., & Konisky, D. M. (2020). The justice and equity implications of the clean energy transition. *Nature Energy*, 5(8), 569–577.
- Chun, J., Ortiz, D., Jin, B., Kulkarni, N., Hart, S., & Knox-Hayes, J. (2025). Energy burden in the United States: An analysis using decision trees. *Energies*, 18(3), 646. <https://doi.org/10.3390/en18030646>
- Ewing, R., & Rong, F. (2008). The impact of urban form on U.S. residential energy use. *Housing Policy Debate*, 19(1), 1–30.
- Güneralp, B., Zhou, Y., Ürge-Vorsatz, D., Gupta, M., Yu, S., Patel, P. L., Fragkias, M., Li, X., & Seto, K. C. (2017). Global scenarios of urban density and its impacts on building energy use through 2050. *PNAS*, 114(34), 8945–8950.
- Hernández, D., & Bird, S. (2010). Energy burden and the need for integrated low-income housing and energy policy. *Poverty & Public Policy*, 2(4), 5–25. <https://doi.org/10.2202/1944-2858.1095>
- Ma, J., & Cheng, J. C. (2016). Identifying the influential features on the regional energy use intensity of residential buildings based on Random Forests. *Applied energy*, 183, 193–201.
- McPherson, E. G., Nowak, D., Heisler, G., Grimmond, S., Souch, C., Grant, R., & Rowntree, R. (1997). Quantifying urban forest structure, function, and value. *Urban Ecosystems*, 1(1), 49–61.
- Melvin, J. (2018). The split incentives energy efficiency problem: Evidence of underinvestment by landlords. *Energy Policy*, 115, 342–352. doi:10.1016/j.enpol.2017.11.069

- New York City Department of Buildings. (2026). Local Law 84 benchmarking law. <https://www.nyc.gov/site/buildings/codes/ll84-benchmarking-law.page>
- New York City Department of Buildings. (2026). Local Law 97 greenhouse gas emissions reductions. <https://www.nyc.gov/site/buildings/codes/ll97-greenhouse-gas-emissions-reductions.page>
- New York City Mayor's Office of Sustainability. (2023). Energy cost burden in New York City. <https://www.nyc.gov/assets/sustainability/downloads/pdf/publications/EnergyCost.pdf>
- New York State Office of Temporary and Disability Assistance. (2026.). Home Energy Assistance Program (HEAP). <https://otda.ny.gov/programs/heap/>
- Papadopoulos, S., Bonczak, B., & Kontokosta, C. E. (2018). Pattern recognition in building energy performance over time using Energy Benchmarking Data. *Applied Energy*, 221, 576–586. <https://doi.org/10.1016/j.apenergy.2018.03.079>
- Ratti, C., Baker, N., & Steemers, K. (2005). Energy consumption and urban texture. *Energy and Buildings*, 37(7), 762–776.
- Shron and Velez (2024). NY HEAT and Energy Affordability in New York City https://switchbox-data.github.io/reports/nyheat_regions/nyc/
- United Nations. (2023). Affordable and clean energy. <https://www.un.org/sustainabledevelopment/energy/>
- Kim, H. S., Yoo, S., Hyeon, J., & Cho, J. (2026). Unmasking indoor exposure profiles in vulnerable households: an unsupervised clustering of integrated environmental and demographic data. *Environment International*, 110167.
- Ko, Y. (2013). Urban form and residential energy use: A review of design principles and research findings. *Journal of planning literature*, 28(4), 327-351. <https://journals.sagepub.com/doi/10.1177/0885412213491499>
- Zhang, Y., Chen, W., Xu, R., & Black, J. (2015). A cluster-based method for calculating baselines for residential loads. *IEEE Transactions on smart grid*, 7(5), 2368-2377. <https://doi.org/10.1109/tsg.2015.2463755>